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Adaptive Active Noise Cancellation System for Small Wind Turbines

ABSTRACT

An adaptive active noise cancellation (ANC) system tailored for small wind turbines in urban settings is disclosed. The system combines an array of directional microphones, a sophisticated multi-core processing unit leveraging machine learning algorithms, and a network of parametric speakers. The invention achieves significant reduction in perceived noise from the turbine across a wide spectrum of wind conditions and ambient acoustic environments, enhancing its suitability for deployment in noise-sensitive areas.

BACKGROUND OF THE INVENTION

The growing demand for renewable energy sources has spurred the development of small wind turbines as a decentralized power generation solution. However, their adoption in densely populated urban areas has been hindered by the noise they produce, which can disturb residents and violate local noise ordinances. Traditional passive noise reduction techniques, such as barriers and enclosures, often compromise turbine efficiency and add complexity to the design. Active noise cancellation offers a promising alternative, but existing systems often struggle to adapt to the dynamic and diverse acoustic conditions found in urban environments.

SUMMARY OF THE INVENTION

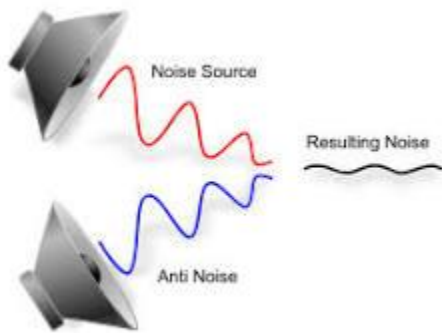
The present invention overcomes these limitations by introducing an adaptive ANC system specifically designed for small wind turbines. The system comprises:

1. **Microphone Array:** A strategically placed array of directional microphones captures the wind turbine's noise signature from multiple angles, providing a comprehensive acoustic input for analysis.
2. **Wind Sensors:** High-precision sensors measure wind speed and direction in real time, enabling the system to anticipate changes in turbine operation and the resulting noise patterns.
3. **Multi-Core Processing Unit:** A powerful multi-core processor, equipped with machine learning algorithms, analyzes the audio input from the microphones and the wind data. It identifies and classifies various noise components (e.g., blade noise, gearbox noise) and predicts their evolution based on the current operating conditions.
4. **Parametric Speaker Network:** A network of parametric speakers, strategically located around the turbine, emits precisely tailored anti-noise signals that destructively interfere with the turbine's noise, effectively canceling it out in the surrounding environment.

The system's adaptive nature is a key innovation. It continuously learns and refines its noise cancellation strategies based on the specific turbine's characteristics and the evolving environmental conditions. This ensures optimal noise reduction performance across a wide range of operating scenarios.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Schematic diagram of the adaptive active noise cancellation system for a small wind turbine.



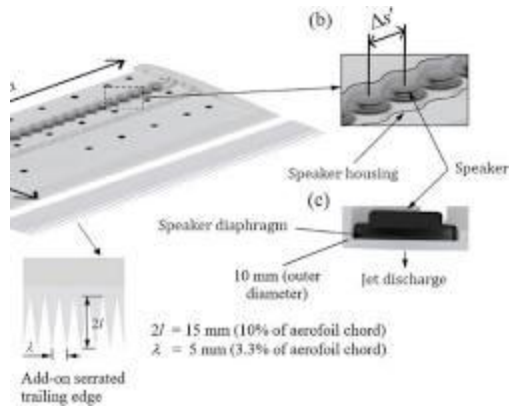
schematic diagram showing a wind turbine with a network of microphones and speakers around it. The microphones are connected to a processing unit, which is also connected to the speakers. The system is labeled Adaptive Active Noise Cancellation System.

FIG. 2: Detailed view of a directional microphone used in the system.



closeup view of a directional microphone mounted on a bracket near the wind turbine.

FIG. 3: Detailed view of a parametric speaker used in the system.



closeup view of a parametric speaker mounted on a bracket near the wind turbine.

FIG. 4: Block diagram of the multi-core processing unit and its components.

DETAILED DESCRIPTION OF THE INVENTION

1. Microphone Array (10):

- Comprises multiple (e.g., 8 or more) directional microphones (12) strategically positioned around the wind turbine (14).
- Each microphone (12) is designed for high sensitivity, wide frequency response, and weather resistance.
- The directional nature of the microphones allows the system to isolate the turbine's noise from other ambient sounds.
- The microphone array (10) provides a spatial distribution of the turbine's noise field, enabling the system to accurately identify and localize noise sources.

2. Wind Sensors (16):

- The system includes at least one wind speed sensor (18) and one wind direction sensor (20).
- These sensors provide real-time data on wind conditions, which are crucial for predicting changes in turbine noise patterns.
- The wind speed sensor (18) measures the velocity of the wind, while the wind direction sensor (20) determines the direction from which the wind is blowing.

3. Multi-Core Processing Unit (22):

- The heart of the system is a powerful multi-core processor (24) with substantial computational capabilities.
- This processor runs sophisticated machine learning algorithms that analyze the audio data from the microphones and the wind data from the sensors.
- The algorithms are trained on a large dataset of wind turbine noise recordings and corresponding wind conditions, enabling them to identify and classify different noise patterns accurately.
- The processing unit (22) also includes a predictive module that uses the analyzed data to forecast how the turbine's noise signature will evolve in the near future.

4. Parametric Speaker Network (26):

- The system utilizes a network of parametric speakers (28) to emit anti-noise signals.
- Parametric speakers generate highly directional ultrasonic waves that, through non-linear interactions in the air, create audible sound waves at the desired frequencies.
- This directionality allows for precise control of the noise cancellation zone, minimizing the impact on the surrounding environment.
- The processing unit (22) determines the optimal amplitude and phase of the anti-noise signals for each speaker (28) based on the current noise conditions and the predicted changes.

CLAIMS

1. An adaptive active noise cancellation system for a wind turbine, comprising:
 - An array of at least eight directional microphones (12) positioned around the wind turbine (14).
 - A multi-core processing unit (22) connected to said microphones (12), configured to receive audio input from said microphones (12).
 - A network of at least six parametric speakers (28) positioned around the wind turbine (14).
 - At least one wind speed sensor (18) and one wind direction sensor (20), configured to provide real-time wind data to said processing unit (22).
 - The processing unit (22) is further configured to:
 - Analyze the audio input using machine learning algorithms to identify and categorize noise patterns generated by the wind turbine (14).
 - Generate cancellation sound waves tailored to the identified noise patterns and current wind conditions.
 - Output the cancellation sound waves through the parametric speakers (28) to reduce the perceived noise of the wind turbine (14).
 - Continuously adjust the cancellation sound waves based on real-time feedback from the microphones (12) and wind sensors (16).
2. The system of claim 1, wherein the machine learning algorithms include a combination of convolutional neural networks for pattern recognition and recurrent neural networks for temporal sequence prediction.
3. The system of claim 1, wherein the parametric speakers (28) use ultrasonic waves to deliver highly directional noise cancellation, minimizing interference with ambient sounds.
4. The system of claim 1, further comprising a database of pre-recorded wind turbine noise patterns, wherein the processing unit (22) uses this database to initialize and enhance its noise cancellation algorithms.

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